# Eduardo Ventilari Sodré

*Curriculum Vitae*

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*evsmath.github.io/*

## Education

2023 – 2025 **Brown University**, *Ph.D. in Mathematics*, Providence, USA

2025 **Brown University**, *M.Sc. in Mathematics*, Providence, USA

2019 – 2022 **Universidade de São Paulo (IME-USP)**, *B.Sc. in Mathematics*, São Paulo, Brazil
Weighted average: 9.8/10, 1st in class

## Publications and Preprints

2026 *Quadratic Hubbard trees, automata, and ray connections*. ArXiv.

2023 *Revisiting Arnold's topological proof of the Morse index theorem*. ArXiv.

2021 *Classification of linear operators satisfying* $(Au, v) = (u, A^r v)$ *or* $(Au, A^r v) = (u, v)$ *on a vector space with indefinite scalar product* (with V. S. Borges, I. Kashyba, V. V. Sergeichuk & A. Zaidan). Linear Algebra and its Applications, v. 611, p. 118-134, 2021.

## Grants and Scholarships

2026 **ICM 2026 Graduate Student and Early Career Travel Grant Award**, *AMS*

2021 – 2022 **FAPESP Undergraduate Research Scholarship**

2019 – 2021 **CNPq Undergraduate Research Scholarship**

2020 **IMPA Summer Program Scholarship**

2019 **PUB Undergraduate Research Scholarship**

## Awards and Fellowships

2024 **Chandler O. Meyer Fellowship Fund in Mathematics**, *Brown University*

2023 **Chancellor Robert Fellow**, *Brown University*

2023 **Institute of Mathematics and Statistics Best Student Prize**, *IME-USP*, Class of 2022

### Olympiads and Competitions

**International Mathematics Competition**, *University College London*
First Prize: 2022, 2021 Second Prize: 2020

**Competencia Iberoamericana Interuniversitaria de Matemáticas**, *Universidad Antonio Nariño*
Gold Medal: 2022 Silver Medal: 2020

**Brazilian Universitary Mathematics Olympiad**, *SBM & IMPA*
Gold Medal: 2022 Silver Medal: 2023, 2020, 2019

**Elon Lages Lima Mathematics Competition**, *SBM & IMPA*
Gold Medal: 2022, 2021 Silver Medal: 2020

## Teaching and Tutoring

Fall 2026 **DRP: Ergodic Theory**

Spring 2026 **Lecturer: "Linear Algebra"**

Fall 2025 **DRP: Beardon's *Iteration of Rational Functions***

Fall 2025 **TA: "Multivariable Calculus"**

Spring 2025 **DRP: Baez and Muniain's *Gauge Fields, Knots and Gravity***

Spring 2025 **TA: "Single Variable Calculus I"**

Fall 2024 **DRP: Lee's *Introduction to Smooth Manifolds***

Fall 2024 **Sheridan Teaching Seminar Program Certificate**, The Harriet W. Sheridan Center for Teaching and Learning

Fall 2024 **TA: "Single Variable Calculus II"**

Spring 2023 **DRP: Devaney's *An Introduction to Discrete Dynamical Systems***

2021 **TA: "Introduction to Complex Analysis"**

## Other Activities

### Positions

2026 - 2027 **President of the Brown Chapter of the AMS**

2024-2025 **Secretary of the Brown Chapter of the AMS**

### Talks

Jun. 2026 **Early Career Complex Dynamics Workshop**, *Hubbard trees, automata and ray connections*, University of Michigan

Jun. 2025 **Topics in Complex Dynamics 2025**, *Thurston Iteration On The Rees Space*, Barcelona

Aug. 2023 **Geometry Research Group Seminar at IME-USP**, *Revisiting Arnold's Topological Proof of the Morse Index Theorem*

### Posters

2025 **JHH80 Dynamical Developments: Degenerations of Flat Surfaces and Rational Maps**, *IMT*
Poster "Thurston Iteration On The Rees Space"

2022 **São Carlos Conference in Geometry**, *ICMC-USP*
Poster "The Sasaki Metric and the Geodesic Flow"

2020 **International Symposium of Scientific Initiation and Technology of USP**, *USP*
Poster "A Generalization of the Theorem on Tangent Bifurcations"

### Seminar Groups

Fall 2025 **Arithmetic, Algebraic & Complex Dynamics Seminar at Brown**, *Brown University*
Co-organizer and Presenter

Spring 2025 **Teichmuller Theory Learning Seminar**, *Brown University*
Organizer and Presenter

**Brown University Geometry/Topology Seminar**, *Brown University*

**Geometry Research Group Seminar at IME-USP**, *IME-USP*

**Quasiconformal Maps and Teichmüller Theory**, *online*

**S4 Student Seminar Group**, *IME-USP*
Organizer and Presenter

**Matrix Problems and Quiver Representations in Linear Algebra**, *IME-USP*

**Aleph 1 Student Seminar on Set Theory**, *IME-USP*

### Events Organized

2024 - 2026 **AMS New England Graduate Student Conference**, *Brown University*

2022 **6th TWAS Young Affiliates Network Thematic Workshop in Mathematics**, *IME-USP*

### Other Events Attended

2026 **ICM**, *Philadelphia*

2026 **Fields Working Group**, *Western University*

2023, 2024 **Geometry and Topology Seminar at Brown and Yale (GATSBY)**, *Brown University*

2023 **34th Brazilian Colloquium of Mathematics (CBM)**, *IMPA*

2022 **Complex Dynamics in the Tropics**, *IMPA*
Financial support by CNPq and FAPERJ

2021 **33rd Brazilian Colloquium of Mathematics (CBM)**, *IMPA*

## Languages

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|---|---|---|
| Portuguese | Native | |
| English | Fluent | TOEFL iBT Score: 118/120, 2022 |
| Spanish | Basic | |
| French | Basic | |